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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 30th January Evening Shift

JEE Main Question Paper with Answer Key

30 Jan 2023	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	22 questions
Physics	26 questions
Out of syllabus	20 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

Formulae for Nessler's reagent is :

(A) $KHgI_3$ (B) HgI_2 (C) KHg_2I_2 (D) K_2HgI_4

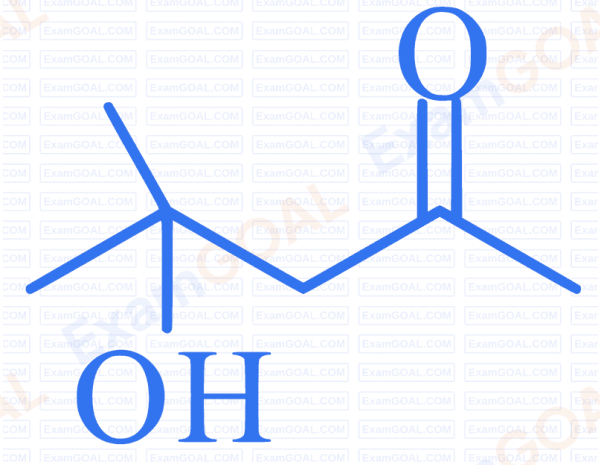
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Q2. MCQ (Single Correct Answer)

Open solution

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A:  can be easily reduced using $Zn - Hg/HCl$ to 

Reason R : $Zn - Hg/HCl$ is used to reduce carbonyl group to $-CH_2-$ group.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) Both A and R are true and R is the correct explanation of A

(B) A is false but R is true

(C) A is true but R is false

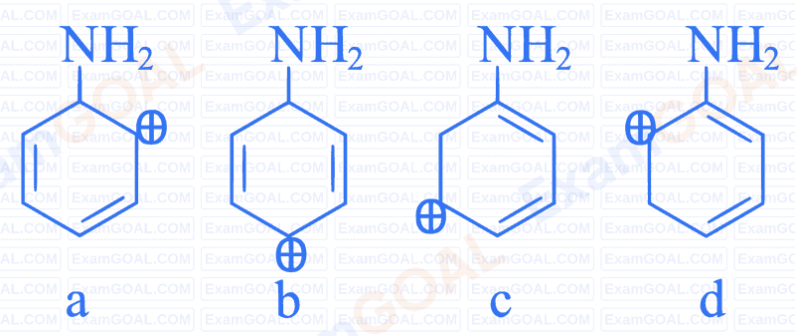
(D) Both A and R are true but R is not the correct explanation of A

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Q3. MCQ (Single Correct Answer)

The most stable carbocation for the following is:

Open solution



(A) b (B) d (C) c (D) a

Q4. MCQ (Single Correct Answer)

Match List **I** with List **II**:

Open solution

List I (Complexes)		List II (Hybridisation)	
A.	$[Ni(CO)_4]$	I.	sp^3
B.	$[Cu(NH_3)_4]^{2+}$	II.	dsp^2
C.	$[Fe(NH_3)_6]^{2+}$	III.	sp^3d^2
D.	$[Fe(H_2O)_6]^{2+}$	IV.	d^2sp^3

(A) A-I, B-II, C-III, D-IV (B) A-II, B-I, C-III, D-IV

(C) A-I, B-II, C-IV, D-III (D) A-II, B-I, C-IV, D-III

Q5. MCQ (Single Correct Answer)

The $Cl - Co - Cl$ bond angle values in a fac- $[Co(NH_3)_3Cl_3]$ complex is/are :

Open solution

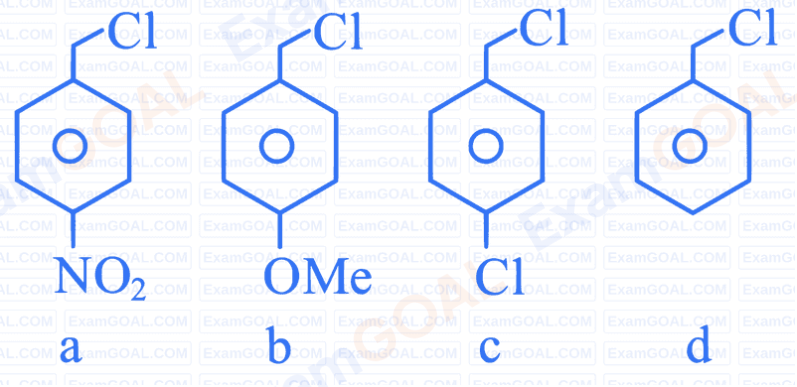
(A) 90° & 120° (B) 90° (C) 180° (D) 90° & 180°

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Q6. MCQ (Single Correct Answer)

Open solution

Decreasing order towards SN 1 reaction for the following compounds is:



(A) $a > c > d > b$

(B) $\mathrm{d} > \mathrm{b} > \mathrm{c} > \mathrm{a}$

(C) $a > b > c > d$

(D) $\mathrm{b} > \mathrm{d} > \mathrm{c} > \mathrm{a}$

Q7. MCQ (Single Correct Answer)

Open solution

$KMnO_4$ oxidises I^- in acidic and neutral/faintly alkaline solutions, respectively, to :

(A) IO_3^- & IO_3^-

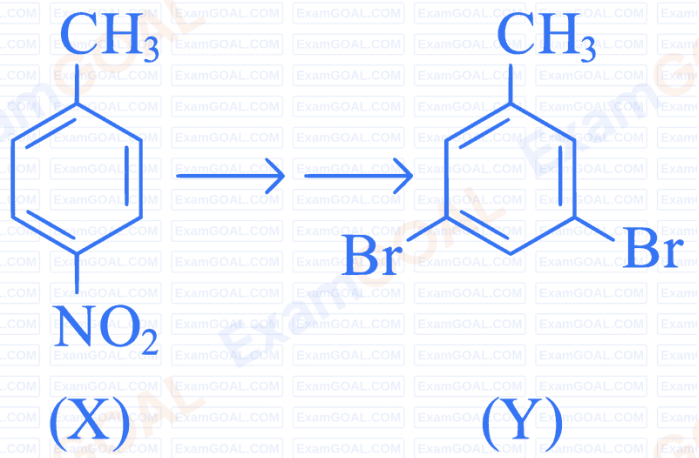
(B) I_2 & I_2

(C) I_2 & IO_3^-

(D) IO_3^- & I_2

Q8. MCQ (Single Correct Answer)

Open solution



In the above conversion of compound (X) to product (Y), the sequence of reagents to be used will be:

(A) (i) Br_2, Fe (ii) Fe, H^+ (iii) $LiAlH_4$

(B) (i) $Br_2(aq)$ (ii) $LiAlH_4$ (iii) H_3O^+

(C) (i) Fe, H^+ (ii) Br_2 (aq) (iii) HNO_2 (iv) H_3PO_2

(D) (i) Fe, H^+ (ii) $Br_2(aq)$ (iii) HNO_2 (iv) $CuBr$

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Q9. MCQ (Single Correct Answer)

Open solution

Match List **I** with List **II**:

List I (Mixture)		List II (Separation Technique)	
A.	$CHCl_3 + C_6H_5NH_2$	I.	Steam distillation
B.	$C_6H_{14} + C_5H_{12}$	II.	Differential extraction
C.	$C_6H_5NH_2 + H_2O$	III.	Distillation
D.	Organic compound in H_2O	IV.	Fractional distillation

(A) A-III, B-I, C-IV, D-II
(B) A-II, B-I, C-III, D-IV
(C) A-IV, B-I, C-III, D-II
(D) A-III, B-IV, C-I, D-II

Q10. MCQ (Single Correct Answer)

Open solution

Maximum number of electrons that can be accommodated in shell with $n = 4$ are:

(A) 50 **(B)** 32 **(C)** 72 **(D)** 16

Q11. MCQ (Single Correct Answer)

Open solution

Bond dissociation energy of "E-H" bond of the "H_2E " hydrides of group 16 elements (given below), follows order.

A. O

B. S

C. Se

D. Te

Choose the correct answer from the options given below:

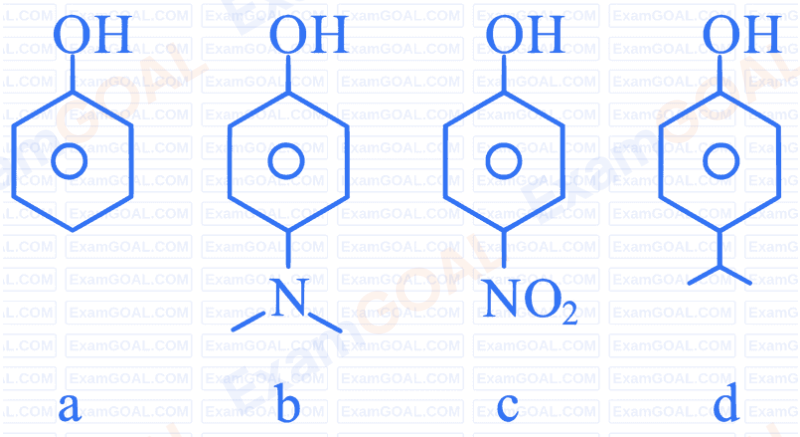
(A) $A > B > C > D$
(B) $D > C > B > A$
(C) $B > A > C > D$
(D) $A > B > D > C$

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Q12. MCQ (Single Correct Answer)

Open solution

The correct order of pK_a values for the following compounds is:



(A) $a > b > c > d$

(B) $c > a > d > b$

(C) $b > d > a > c$

(D) $b > a > d > c$

Q13. MCQ (Single Correct Answer)

Open solution

$1\,L, 0.02M$ solution of $[Co(NH_3)_5SO_4]$ Br is mixed with $1\,L, 0.02M$ solution of $[Co(NH_3)_5Br]SO_4$. The resulting solution is divided into two equal parts (X) and treated with excess of $AgNO_3$ solution and $BaCl_2$ solution respectively as shown below:

$1\,L$ Solution $(X) + AgNO_3$ solution (excess) $\longrightarrow Y$

$1\,L$ Solution $(X) + BaCl_2$ solution (excess) $\longrightarrow Z$

The number of moles of Y and Z respectively are

(A) $0.01, 0.01$

(B) $0.01, 0.02$

(C) $0.02, 0.01$

(D) $0.02, 0.02$

Q14. MCQ (Single Correct Answer)

Open solution

The wave function (Ψ) of $2\,s$ is given by

$$\Psi_{2\,s} = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{1/2}\left(2 - \frac{r}{a_0}\right)e^{-r/2a_0}$$

At $r = r_0$, radial node is formed. Thus, r_0 in terms of a_0

(A) $r_0 = a_0$

(B) $r_0 = 4a_0$

(C) $r_0 = 2a_0$

(D) $r_0 = \frac{a_0}{2}$

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Q15. Numerical

The strength of 50 volume solution of hydrogen peroxide is _____ $\mathrm{g/L}$ (Nearest integer).

Given:

Molar mass of $\mathrm{H_2O_2}$ is $34\ \mathrm{g\ mol^{-1}}$

Molar volume of gas at $\mathrm{STP} = 22.7\ \mathrm{L}$

Open solution

Q16. Numerical

A short peptide on complete hydrolysis produces 3 moles of glycine (G), two moles of leucine (L) and two moles of valine (V) per mole of peptide. The number of peptide linkages in it are _____.

Open solution

Q17. Numerical

Number of compounds from the following which will not dissolve in cold $\mathrm{NaHCO_3}$ and NaOH solutions but will dissolve in hot NaOH solution is _____.



Open solution

Q18. Numerical

An organic compound undergoes first-order decomposition. If the time taken for the 60% decomposition is $540\ \mathrm{s}$, then the time required for 90% decomposition will be _____ s. (Nearest integer).

Given: $\ln 10 = 2.3; \log 2 = 0.3$

Open solution

Q19. Numerical

1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of $27^{\circ}\mathrm{C}$. The work done is $3\ \mathrm{kJ\ mol^{-1}}$. The final temperature of the gas is _____ K (Nearest integer).

Given $\mathrm{C_V} = 20\ \mathrm{J\ mol^{-1}\ K^{-1}}$

Open solution

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Q20. Numerical

The electrode potential of the following half cell at $298\ \mathrm{K}$

$\mathrm{X}\left|\mathrm{X}^{2+}(0.001\mathrm{M})\right\|\mathrm{Y}^{2+}(0.01\mathrm{M})\left|\mathrm{Y}\right.$ is _____ $\times 10^{-2}\ \mathrm{V}$ (Nearest integer)

Open solution

Given: $\mathrm{E}^{0}_{\mathrm{X}^{2+}|\mathrm{X}} = -2.36\ \mathrm{V}$

$\mathrm{E}^{0}_{\mathrm{Y}^{2+}|\mathrm{Y}} = +0.36\ \mathrm{V}$

$\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.06\ \mathrm{V}$

Q21. Numerical

Consider the following equation:

$2\mathrm{SO}_2(g) + \mathrm{O}_2(g) \rightleftharpoons 2\mathrm{SO}_3(g), \Delta H = -190\ \mathrm{kJ}$

Open solution

The number of factors which will increase the yield of SO_3 at equilibrium from the following is _____.

A. Increasing temperature

B. Increasing pressure

C. Adding more SO_2

D. Adding more O_2

E. Addition of catalyst

Q22. Numerical

Lead storage battery contains 38% by weight solution of $\mathrm{H}_2\mathrm{SO}_4$. The van't Hoff factor is 2.67 at this concentration. The temperature in Kelvin at which the solution in the battery will freeze is _____. (Nearest integer).

Open solution

Given $\mathrm{K}_f = 1.8\ \mathrm{K\ kg\ mol}^{-1}$

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Mathematics 22 questions

Q23. MCQ (More than One Correct Answer)

Let $a_1 = 1, a_2, a_3, a_4, \ldots..$ be consecutive natural numbers.

Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{1}{1+a_2a_3}\right) + \ldots.. + \tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to :

(A) $\frac{\pi}{4} - \cot^{-1}(2022)$ (B) $\frac{\pi}{4} - \tan^{-1}(2022)$ (C) $\cot^{-1}(2022) - \frac{\pi}{4}$ (D) $\tan^{-1}(2022) - \frac{\pi}{4}$

Open solution

Q24. MCQ (Single Correct Answer)

For $\alpha, \beta \in \mathbb{R}$, suppose the system of linear equations

$x - y + z = 5$
$2x + 2y + \alpha z = 8$
$3x - y + 4z = \beta$

has infinitely many solutions. Then α and β are the roots of :

(A) $x^2 + 18x + 56 = 0$ (B) $x^2 - 10x + 16 = 0$ (C) $x^2 + 14x + 24 = 0$ (D) $x^2 - 18x + 56 = 0$

Open solution

Q25. MCQ (Single Correct Answer)

The number of ways of selecting two numbers a and $b, a \in \{2, 4, 6, \ldots., 100\}$ and $b \in \{1, 3, 5, \ldots.., 99\}$ such that 2 is the remainder when $a + b$ is divided by 23 is :

(A) 186 (B) 54 (C) 108 (D) 268

Open solution

Q26. MCQ (Single Correct Answer)

Let S be the set of all values of a_1 for which the mean deviation about the mean of 100 consecutive positive integers $a_1, a_2, a_3, \ldots., a_{100}$ is 25 . Then S is :

(A) $\{9\}$ (B) ϕ (C) $\{99\}$ (D) N

Open solution

Q27. MCQ (Single Correct Answer)

Let $a, b, c > 1, a^3, b^3$ and c^3 be in A.P., and $\log_a b, \log_c a$ and $\log_b c$ be in G.P. If the sum of first 20 terms of an A.P., whose first term is $\frac{a+4b+c}{3}$ and the common difference is $\frac{a-8b+c}{10}$ is -444, then abc is equal to :

(A) 343 (B) 216 (C) $\frac{343}{8}$ (D) $\frac{125}{8}$

Open solution

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Q28. MCQ (Single Correct Answer)

Open solution

Let f, g and h be the real valued functions defined on $\mathbb{R}$ as

$$f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

$$g(x) = \begin{cases} \frac{\sin(x+1)}{(x+1)}, & x \neq -1 \\ 1, & x = -1 \end{cases}$$

and $h(x) = 2[x] - f(x)$, where $[x]$ is the greatest integer $\leq x$. Then the

value of $\lim\limits_{x \to 1} g(h(x-1))$ is :

(A) 1 (B) -1 (C) $\sin(1)$ (D) 0

Q29. MCQ (Single Correct Answer)

Open solution

Let q be the maximum integral value of p in $[0, 10]$ for which the roots of the equation $x^2 - px + \frac{5}{4}p = 0$ are rational. Then the area of the region $\{(x, y) : 0 \leq y \leq (x - q)^2, 0 \leq x \leq q\}$ is :

(A) $\frac{125}{3}$ (B) 243 (C) 164 (D) 25

Q30. MCQ (Single Correct Answer)

Open solution

If the functions $f(x) = \frac{x^3}{3} + 2bx + \frac{ax^2}{2}$

and $g(x) = \frac{x^3}{3} + ax + bx^2, a \neq 2b$

have a common extreme point, then $a + 2b + 7$ is equal to :

(A) 6 (B) $\frac{3}{2}$ (C) 3 (D) 4

Q31. MCQ (Single Correct Answer)

Open solution

The parabolas : $ax^2 + 2bx + cy = 0$ and $dx^2 + 2ex + fy = 0$ intersect on the line $y = 1$. If a, b, c, d, e, f are positive real numbers and a, b, c are in G.P., then :

(A) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (B) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(C) d, e, f are in A.P. (D) d, e, f are in G.P.

Q32. MCQ (Single Correct Answer)

Open solution

Let $\vec{a}$ and $\vec{b}$ be two vectors, Let $|\vec{a}| = 1, |\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$, then the value of $\vec{b} \cdot \vec{c}$ is :

(A) -48 (B) -60 (C) -84 (D) -24

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Q33. MCQ (Single Correct Answer)

The range of the function $f(x) = \sqrt{3-x} + \sqrt{2+x}$ is :

(A) $[2\sqrt{2}, \sqrt{11}]$ (B) $[\sqrt{5}, \sqrt{13}]$ (C) $[\sqrt{2}, \sqrt{7}]$ (D) $[\sqrt{5}, \sqrt{10}]$

Open solution

Q34. MCQ (Single Correct Answer)

The solution of the differential equation

$\frac{dy}{dx} = -\left(\frac{x^2+3y^2}{3x^2+y^2}\right), y(1) = 0$ is :

(A) $\log_e|x+y| + \frac{xy}{(x+y)^2} = 0$ (B) $\log_e|x+y| - \frac{xy}{(x+y)^2} = 0$

(C) $\log_e|x+y| + \frac{2xy}{(x+y)^2} = 0$ (D) $\log_e|x+y| - \frac{2xy}{(x+y)^2} = 0$

Open solution

Q35. MCQ (Single Correct Answer)

Let $x = (8\sqrt{3}+13)^{13}$ and $y = (7\sqrt{2}+9)^9$. If $[t]$ denotes the greatest integer $\leq t$, then :

(A) $[x]$ is odd but $[y]$ is even (B) $[x]$ and $[y]$ are both odd

(C) $[x] + [y]$ is even (D) $[x]$ is even but $[y]$ is odd

Open solution

Q36. Numerical

Let $A = \{1, 2, 3, 5, 8, 9\}$. Then the number of possible functions $f : A \to A$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in A$ with $m \cdot n \in A$ is equal to _____.

Open solution

Q37. Numerical

The number of seven digits odd numbers, that can be formed using all the seven digits 1, 2, 2, 2, 3, 3, 5 is _____.

Open solution

Q38. Numerical

Let A be the area of the region

$\{(x, y) : y \geq x^2, y \geq (1-x)^2, y \leq 2x(1-x)\}$.

Then $540\ A$ is equal to :

Open solution

Q39. Numerical

If the value of real number $a > 0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$ have a common real root is $\frac{3}{\sqrt{2\beta}}$ then β is equal to _____.

Open solution

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Q40. Numerical

Let $P(a_1, b_1)$ and $Q(a_2, b_2)$ be two distinct points on a circle with center $C(\sqrt{2}, \sqrt{3})$. Let O be the origin and OC be perpendicular to both CP and CQ. If the area of the triangle OCP is $\frac{\sqrt{35}}{2}$, then $a_1^2 + a_2^2 + b_1^2 + b_2^2$ is equal to :

Open solution

Q41. Numerical

Let a line L pass through the point $P(2, 3, 1)$ and be parallel to the line $x + 3y - 2z - 2 = 0 = x - y + 2z$. If the distance of L from the point $(5, 3, 8)$ is α, then $3\alpha^2$ is equal to :

Open solution

Q42. Numerical

A bag contains six balls of different colours. Two balls are drawn in succession with replacement. The probability that both the balls are of the same colour is p. Next four balls are drawn in succession with replacement and the probability that exactly three balls are of the same colour is q. If $p : q = m : n$, where m and n are coprime, then $m + n$ is equal to :

Open solution

Q43. Numerical

If $\int \sqrt{\sec 2x - 1}dx = \alpha \log_e \left|\cos 2x + \beta + \sqrt{\cos 2x \left(1 + \cos \frac{1}{\beta}x\right)}\right| + \text{constant}$, then $\beta - \alpha$ is equal to _____.

Open solution

Q44. Numerical

50th root of a number x is 12 and 50th root of another number y is 18 . Then the remainder obtained on dividing $(x + y)$ by 25 is _____.

Open solution

Physics 26 questions

Q45. MCQ (Single Correct Answer)

A vehicle travels $4\ \text{km}$ with speed of $3\ \text{km/h}$ and another $4\ \text{km}$ with speed of $5\ \text{km/h}$, then its average speed is

(A) $3.75\ \text{km/h}$ (B) $4.25\ \text{km/h}$ (C) $3.50\ \text{km/h}$ (D) $4.00\ \text{km/h}$

Open solution

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Q46. MCQ (Single Correct Answer)

Open solution

Match List **I** with List **II**:

List I		List II	
A.	Torque	I.	$\mathrm{kgm^{-1}\,s^{-2}}$
B.	Energy density	II.	$\mathrm{kg\,ms^{-1}}$
C.	Pressure gradient	III.	$\mathrm{kg\,m^{-2}\,s^{-2}}$
D.	Impulse	IV.	$\mathrm{kg\,m^{2}\,s^{-2}}$

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II

(B) A-IV, B-I, C-II, D-III

(C) A-I, B-IV, C-III, D-II

(D) A-IV, B-III, C-I, D-II

Q47. MCQ (Single Correct Answer)

Open solution

An object is allowed to fall from a height R above the earth, where R is the radius of earth. Its velocity when it strikes the earth's surface, ignoring air resistance, will be

(A) $\sqrt{\frac{gR}{2}}$

(B) $\sqrt{gR}$

(C) $\sqrt{2gR}$

(D) $2\sqrt{gR}$

Q48. MCQ (Single Correct Answer)

Open solution

A force is applied to a steel wire 'A', rigidly clamped at one end. As a result elongation in the wire is $0.2\ \mathrm{mm}$. If same force is applied to another steel wire ' B ' of double the length and a diameter 2.4 times that of the wire ' A ', the elongation in the wire ' B ' will be (wires having uniform circular cross sections)

(A) $6.9 \times 10^{-2}\ \mathrm{mm}$

(B) $6.06 \times 10^{-2}\ \mathrm{mm}$

(C) $2.77 \times 10^{-2}\ \mathrm{mm}$

(D) $3.0 \times 10^{-2}\ \mathrm{mm}$

Q49. MCQ (Single Correct Answer)

Open solution

An electron accelerated through a potential difference V_1 has a de-Broglie wavelength of λ. When the potential is changed to V_2, its de-Broglie wavelength increases by 50%. The value of $\left(\frac{V_1}{V_2}\right)$ is equal to

(A) $\frac{3}{2}$

(B) 4

(C) 3

(D) $\frac{9}{4}$

Q50. MCQ (Single Correct Answer)

Open solution

A machine gun of mass $10\ \mathrm{kg}$ fires $20\ \mathrm{g}$ bullets at the rate of 180 bullets per minute with a speed of $100\ \mathrm{m\,s^{-1}}$ each. The recoil velocity of the gun is

(A) $0.02\ \mathrm{m/s}$

(B) $1.5\ \mathrm{m/s}$

(C) $2.5\ \mathrm{m/s}$

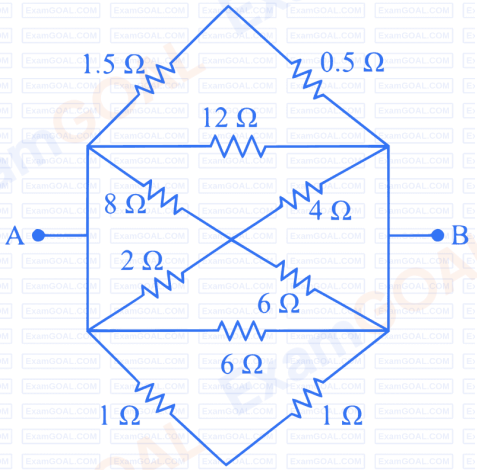
(D) $0.6\ \mathrm{m/s}$

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Q51. MCQ (Single Correct Answer)

The equivalent resistance between A and B is _____.

Open solution

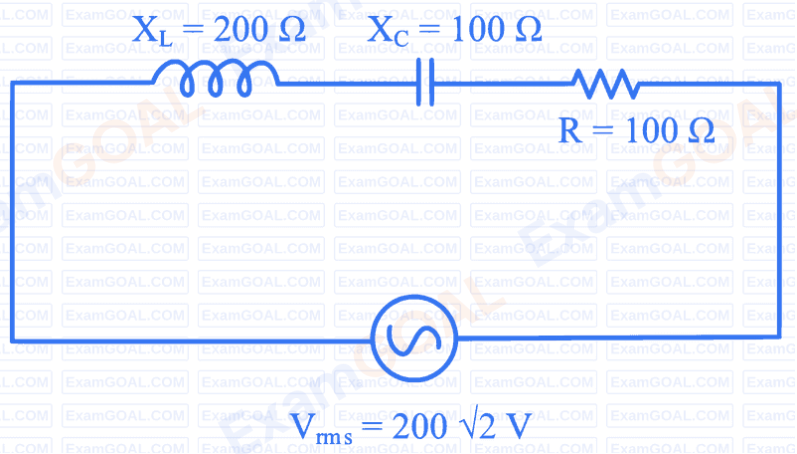


(A) $\frac{1}{2}\Omega$ (B) $\frac{2}{3}\Omega$ (C) $\frac{3}{2}\Omega$ (D) $\frac{1}{3}\Omega$

Q52. MCQ (Single Correct Answer)

In the given circuit, rms value of current (I_{rms}) through the resistor R is:

Open solution



(A) $2\sqrt{2}\,\text{A}$ (B) $2\,\text{A}$ (C) $\frac{1}{2}\,\text{A}$ (D) $20\,\text{A}$

Q53. MCQ (Single Correct Answer)

A thin prism P_1 with an angle 6° and made of glass of refractive index 1.54 is combined with another prism P_2 made from glass of refractive index 1.72 to produce dispersion without average deviation. The angle of prism P_2 is

Open solution

(A) 4.5° (B) 7.8° (C) 1.3° (D) 6°

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Q54. MCQ (Single Correct Answer)

A point source of $100\ \mathrm{W}$ emits light with 5% efficiency. At a distance of $5\ \mathrm{m}$ from the source, the intensity produced by the electric field component is:

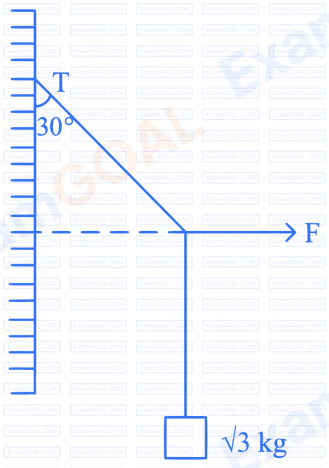
(A) $\frac{1}{40\pi}\frac{W}{m^2}$ (B) $\frac{1}{10\pi}\frac{W}{m^2}$ (C) $\frac{1}{20\pi}\frac{W}{m^2}$ (D) $\frac{1}{2\pi}\frac{W}{m^2}$

Open solution

Q55. MCQ (Single Correct Answer)

A block of $\sqrt{3}\ \mathrm{kg}$ is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30° with the wall. The tension T is: (Given $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)

Open solution



(A) 15 N (B) 10 N (C) 25 N (D) 20 N

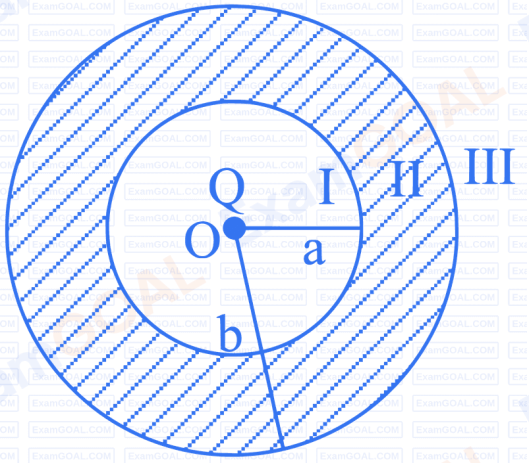
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Q56. MCQ (Single Correct Answer)

Open solution

As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b. The electric field due to charge Q in three different regions I, II and III is given by:

$(\mathrm{I}: r < a, \mathrm{II}: a < r < b, \mathrm{III}: r > b)$



(A) $E_I = 0, E_{II} = 0, E_{III} \neq 0$

(B) $E_I \neq 0, E_{II} = 0, E_{III} = 0$

(C) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$

(D) $E_I = 0, E_{II} = 0, E_{III} = 0$

Q57. MCQ (Single Correct Answer)

Open solution

A flask contains hydrogen and oxygen in the ratio of $2:1$ by mass at temperature $27^{\circ}\mathrm{C}$. The ratio of average kinetic energy per molecule of hydrogen and oxygen respectively is:

(A) 1 : 1

(B) 4 : 1

(C) 1 : 4

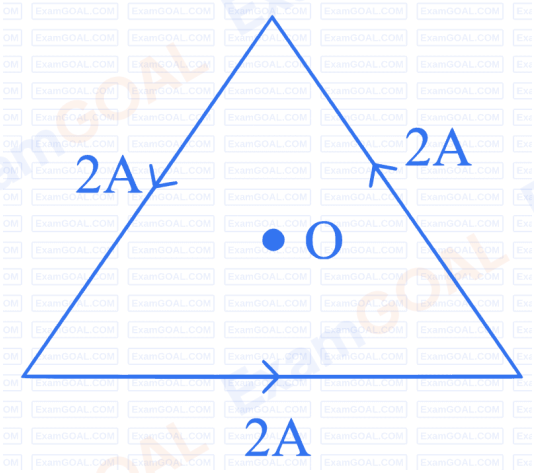
(D) 2 : 1

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Q58. MCQ (Single Correct Answer)

Open solution

As shown in the figure, a current of $2\ \mathrm{A}$ flowing in an equilateral triangle of side $4\sqrt{3}\ \mathrm{cm}$. The magnetic field at the centroid O of the triangle is



(Neglect the effect of earth's magnetic field)

(A) $4\sqrt{3} \times 10^{-4}\ \mathrm{T}$

(B) $4\sqrt{3} \times 10^{-5}\ \mathrm{T}$

(C) $3\sqrt{3} \times 10^{-5}\ \mathrm{T}$

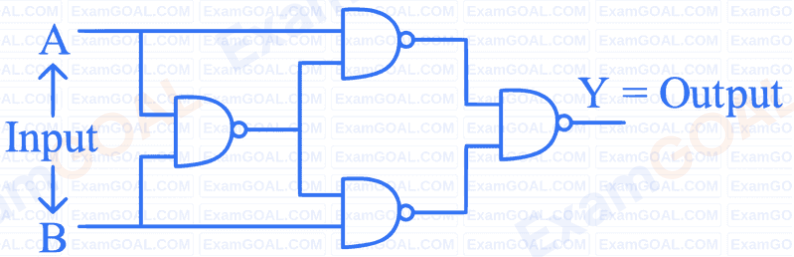
(D) $\sqrt{3} \times 10^{-4}\ \mathrm{T}$

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Q59. MCQ (Single Correct Answer)

Open solution

The output Y for the inputs A and B of circuit is given by



Truth table of the shown circuit is:

(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(D)

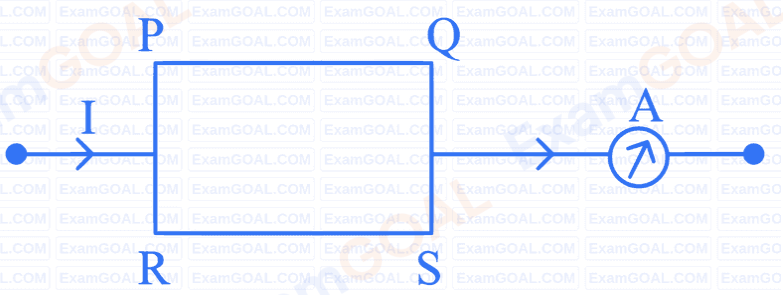
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

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Q60. MCQ (Single Correct Answer)

Open solution

A current carrying rectangular loop PQRS is made of uniform wire. The length $PR = QS = 5\ \text{cm}$ and $PQ = RS = 100\ \text{cm}$. If ammeter current reading changes from I to $2I$, the ratio of magnetic forces per unit length on the wire PQ due to wire RS in the two cases respectively $\left(f_{PQ}^{I} : f_{PQ}^{2I}\right)$ is:

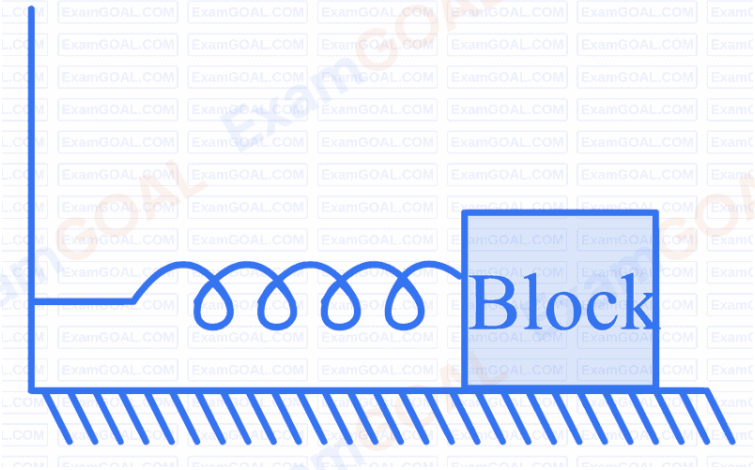


(A) 1 : 4 (B) 1 : 3 (C) 1 : 2 (D) 1 : 5

Q61. MCQ (Single Correct Answer)

Open solution

For a simple harmonic motion in a mass spring system shown, the surface is frictionless. When the mass of the block is $1\ \text{kg}$, the angular frequency is ω_1. When the mass block is $2\ \text{kg}$ the angular frequency is ω_2. The ratio ω_2/ω_1 is



(A) $1/\sqrt{2}$ (B) $1/2$ (C) 2 (D) $\sqrt{2}$

Q62. Numerical

Open solution

In an ac generator, a rectangular coil of 100 turns each having area $14 \times 10^{-2}\ \text{m}^2$ is rotated at $360\ \text{rev/min}$ about an axis perpendicular to a uniform magnetic field of magnitude $3.0\ \text{T}$. The maximum value of the emf produced will be _____ V.

(Take $\pi = \frac{22}{7}$)

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Q63. Numerical

A faulty thermometer reads $5^{\circ}\mathrm{C}$ in melting ice and $95^{\circ}\mathrm{C}$ in steam. The correct temperature on absolute scale will be _____ K when the faulty thermometer reads $41^{\circ}\mathrm{C}$.

Open solution

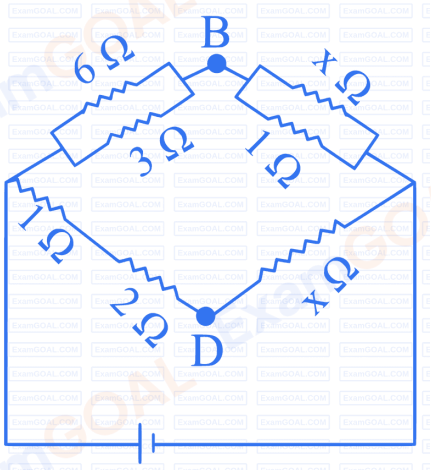
Q64. Numerical

In a Young's double slit experiment, the intensities at two points, for the path differences $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1+I_2}{I_0}$ = _____.

Open solution

Q65. Numerical

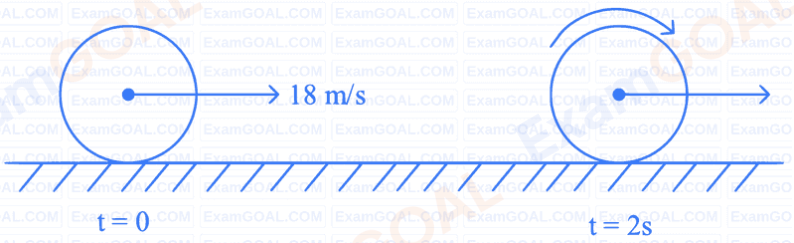
If the potential difference between B and D is zero, the value of x is $\frac{1}{n}\Omega$. The value of n is _____.



Open solution

Q66. Numerical

A uniform disc of mass $0.5\ \mathrm{kg}$ and radius r is projected with velocity $18\ \mathrm{m/s}$ at $\mathrm{t} = 0$ s on a rough horizontal surface. It starts off with a purely sliding motion at $\mathrm{t} = 0$ s. After 2 s it acquires a purely rolling motion (see figure). The total kinetic energy of the disc after 2 s will be _____ J (given, coefficient of friction is 0.3 and $g = 10\ \mathrm{m/s^2}$).



Open solution

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Q67. Numerical

The velocity of a particle executing SHM varies with displacement (x) as $4v^2 = 50 - x^2$. The time period of oscillations is $\frac{x}{7}s$. The value of x is _____. (Take $\pi = \frac{22}{7}$)

Open solution

Q68. Numerical

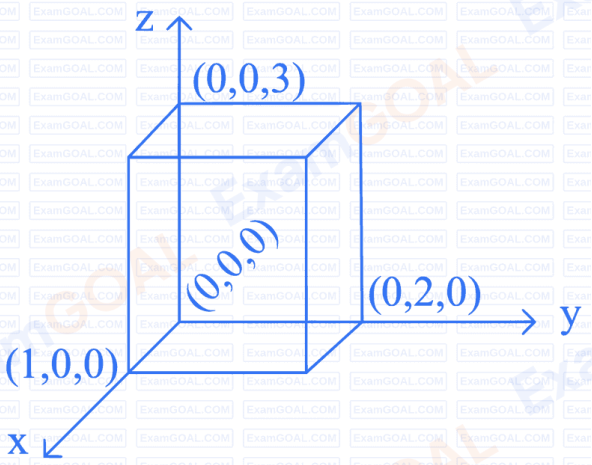
A body of mass $2\,\text{kg}$ is initially at rest. It starts moving unidirectionally under the influence of a source of constant power P. Its displacement in $4\,\text{s}$ is $\frac{1}{3}\alpha^2\sqrt{P}\,\text{m}$. The value of α will be _____.

Open solution

Q69. Numerical

As shown in figure, a cuboid lies in a region with electric field $E = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}\,\text{N/C}$. The magnitude of charge within the cuboid is $n \in_0 C$.

The value of n is _____ (if dimension of cuboid is $1 \times 2 \times 3\,\text{m}^3$)



Q70. Numerical

A stone tied to $180\,\text{cm}$ long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x}ms^{-2}$. The value of x _____. (Take $\pi = \frac{22}{7}$)

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements:

Statement I: During Electrolytic refining, the pure metal is made to act as anode and its impure metallic form is used as cathode.

Statement II: During the Hall-Heroult electrolysis process, purified Al_2O_3 is mixed with Na_3AlF_6 to lower the melting point of the mixture.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Both Statement I and Statement II are correct

Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The water quality of a pond was analyzed and its BOD was found to be 4. The pond has :

(A) Very clean water

(B) Highly polluted water

(C) Water has high amount of fluoride compounds

(D) Slightly polluted water

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Chlorides of which metal are soluble in organic solvents :

(A) K (B) Be (C) Ca (D) Mg

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which of the following reaction is correct?

(A) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$

(B) $2LiNO_3 \longrightarrow 2Li + 2NO_2 + O_2$

(C) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 2\,N_2O_4 + O_2$

(D) $2LiNO_3 \xrightarrow{\Delta} 2LiNO_2 + O_2$

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Q75. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Antihistamines do not affect the secretion of acid in stomach.

Reason R : Antiallergic and antacid drugs work on different receptors.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) A is false but R is true

(B) Both A and R are true but R is not the correct explanation of A

(C) Both A and R are true and R is the correct explanation of A

(D) A is true but R is false

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Boric acid is solid, whereas BF_3 is gas at room temperature because of :

(A) Strong van der Waal's interaction in Boric acid

(B) Strong covalent bond in BF_3

(C) Strong hydrogen bond in Boric acid

(D) Strong ionic bond in Boric acid

Open solution

Q77. Numerical · **Chemistry**

The graph of $\log \frac{x}{m}$ vs $\log p$ for an adsorption process is a straight line inclined at an angle of 45° with intercept equal to 0.6020. The mass of gas adsorbed per unit mass of adsorbent at the pressure of $0.4\ atm$ is _____ $\times 10^{-1}$ (Nearest integer).

Given: $\log 2 = 0.3010$

Open solution

Q78. Numerical · **Chemistry**

Iron oxide FeO, crystallises in a cubic lattice with a unit cell edge length of 5.0 Å. If density of the FeO in the crystal is $4.0\ g\ cm^{-3}$, then the number of FeO units present per unit cell is _____. (Nearest integer)

Given: Molar mass of Fe and O is 56 and $16\ g\ mol^{-1}$ respectively. $N_A = 6.0 \times 10^{23}\ mol^{-1}$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

A vector $\vec{v}$ in the first octant is inclined to the x-axis at 60°, to the y-axis at 45 and to the z-axis at an acute angle. If a plane passing through the points $(\sqrt{2}, -1, 1)$ and (a, b, c), is normal to $\vec{v}$, then :

(A) $a + b + \sqrt{2}c = 1$ (B) $\sqrt{2}a + b + c = 1$ (C) $\sqrt{2}a - b + c = 1$ (D) $a + \sqrt{2}b + c = 1$

Open solution

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Q80. MCQ (Single Correct Answer) · **Mathematics**

If a plane passes through the points $(-1, k, 0), (2, k, -1), (1, 1, 2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ is :

(A) $\frac{17}{5}$ (B) $\frac{6}{13}$ (C) $\frac{13}{6}$ (D) $\frac{5}{17}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

$\lim\limits_{n\to\infty} \frac{3}{n}\left\{4 + \left(2+\frac{1}{n}\right)^2 + \left(2+\frac{2}{n}\right)^2 + \ldots + \left(3-\frac{1}{n}\right)^2\right\}$ is equal to :

(A) 0 (B) $\frac{19}{3}$ (C) 19 (D) 12

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Consider the following statements:

P : I have fever

Q: I will not take medicine

R : I will take rest.

The statement "If I have fever, then I will take medicine and I will take rest" is equivalent to :

(A) $((\sim P) \vee \sim Q) \wedge ((\sim P) \vee \sim R)$ (B) $(P \vee \sim Q) \wedge (P \vee \sim R)$

(C) $((\sim P) \vee \sim Q) \wedge ((\sim P) \vee R)$ (D) $(P \vee Q) \wedge ((\sim P) \vee R)$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let A be a point on the x-axis. Common tangents are drawn from A to the curves $x^2 + y^2 = 8$ and $y^2 = 16x$. If one of these tangents touches the two curves at Q and R, then $(QR)^2$ is equal to :

(A) 76 (B) 81 (C) 64 (D) 72

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let $\lambda \in \mathbb{R}, \vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}, \vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$.

If $((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$,

then $|\lambda(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})|^2$ is equal to :

(A) 136 (B) 140 (C) 144 (D) 132

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

If P is a 3×3 real matrix such that $P^T = aP + (a-1)I$, where $a > 1$, then :

(A) $|AdjP| = 1$ (B) $|AdjP| > 1$

(C) $|AdjP| = \frac{1}{2}$ (D) P is a singular matrix

Open solution

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Q86. Numerical · **Mathematics**

Open solution

The 8th common term of the series

$S_1 = 3 + 7 + 11 + 15 + 19 + \ldots\ldots$

$S_2 = 1 + 6 + 11 + 16 + 21 + \ldots\ldots$

is :

Q87. MCQ (Single Correct Answer) · **Physics**

Open solution

Match List **I** with List **II**:

List I	List II
A. Attenuation	I. Combination of a receiver and transmitter.
B. Transducer	II. process of retrieval of information from the carrier wave at receiver
C. Demodulation	III. converts one form of energy into another
D. quad Repeater	IV. Loss of strength of a signal while propagating through a medium.

Choose the correct answer from the options given below:

(A) A-I, B-II, C-III, D-IV

(B) A-IV, B-III, C-I, D-II

(C) A-IV, B-III, C-II, D-I

(D) A-II, B-III, C-IV, D-I

Q88. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: The nuclear density of nuclides ${}^{10}_{5}\mathrm{B}$, ${}^{6}_{3}\mathrm{Li}$, ${}^{56}_{26}\mathrm{Fe}$, ${}^{20}_{10}\mathrm{Ne}$ and ${}^{209}_{83}\mathrm{Bi}$ can be arranged as $\rho^{\mathrm{N}}_{\mathrm{Bi}} > \rho^{\mathrm{N}}_{\mathrm{Fe}} > \rho^{\mathrm{N}}_{\mathrm{Ne}} > \rho^{\mathrm{N}}_{\mathrm{B}} > \rho^{\mathrm{N}}_{\mathrm{Li}}$

Reason R: The radius R of nucleus is related to its mass number A as $R = R_0 A^{1/3}$, where R_0 is a constant.

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

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Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: Efficiency of a reversible heat engine will be highest at $-273^{\circ}\mathrm{C}$ temperature of cold reservoir.

Reason R: The efficiency of Carnot's engine depends not only on the temperature of the cold reservoir but it depends on the temperature of the hot reservoir too and is given as $\eta = \left(1 - \frac{T_2}{T_1}\right)$

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is false but $\mathbf{R}$ is true

(D) A is true but $\mathbf{R}$ is false

Q90. Numerical · **Physics**

Open solution

A radioactive nucleus decays by two different processes. The half life of the first process is 5 minutes and that of the second process is $30\ \mathrm{s}$. The effective half-life of the nucleus is calculated to be $\frac{\alpha}{11}\ \mathrm{s}$. The value of α is _____.

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Answer Key

Chemistry

1. **D**	2. **B**	3. **C**	4. **C**	5. **B**
6. **D**	7. **C**	8. **C**	9. **D**	10. **B**
11. **A**	12. **C**	13. **A**	14. **C**	15. **150**
16. **6**	17. **3**	18. **1350**	19. **150**	20. **275**
21. **3**	22. **243**			

Mathematics

23. **A, D**	24. **D**	25. **C**	26. **D**	27. **B**
28. **A**	29. **B**	30. **A**	31. **A**	32. **A**
33. **D**	34. **C**	35. **C**	36. **432**	37. **240**
38. **25**	39. **13**	40. **24**	41. **158**	42. **14**
43. **1**	44. **23**			

Physics

45. **A**	46. **A**	47. **B**	48. **A**	49. **D**
50. **D**	51. **B**	52. **B**	53. **A**	54. **A**
55. **D**	56. **C**	57. **A**	58. **C**	59. **A**
60. **A**	61. **A**	62. **1584**	63. **313**	64. **3**
65. **2**	66. **54**	67. **88**	68. **4**	69. **12**
70. **125**				

Out of syllabus

71. **B**	72. **A**	73. **B**	74. **A**	75. **C**
76. **C**	77. **16**	78. **4**	79. **D**	80. **C**
81. **C**	82. **C**	83. **D**	84. **B**	85. **A**
86. **151**	87. **C**	88. **C**	89. **A**	90. **300**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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